The Maturation of MAAT

MAAT Structural Selection Series — Paper 36

From Projection Observables (v0.11)
to Structural Selection (v1.0)
to Emergent Robustness Closure (v1.2.1)

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Abstract

The MAAT framework has evolved from a structural diagnostic tool into a coherent effective theory with a closed interpretation of robustness. This paper reconstructs the conceptual progression through three stages: the projection observable layer of v0.11, the structural selection principle of v1.0, and the emergent robustness closure of v1.2.1.

The central result is that this evolution is *continuous and internally consistent*: each stage resolves a limitation of the previous one without modifying the selection functional or observable pipeline. Projection diagnostics motivate a selection principle; the selection principle reveals an ambiguity in the robustness sector; that ambiguity is resolved by reconstructing robustness as an emergent property of the coherence fields.

The final framework is not more complex than v0.11 — it is more closed. Robustness is no longer an independent input but a consequence of structural coherence, balance, and connectedness.

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1 Introduction

The MAAT programme investigates structural selection in nonlinear field systems and cosmology [17]. Rather than proposing a complete theory in a single step, the framework has developed through a sequence of papers, each one refining the conceptual and mathematical foundations of the previous stage.

This paper traces that development as a single coherent argument. The three stages are:

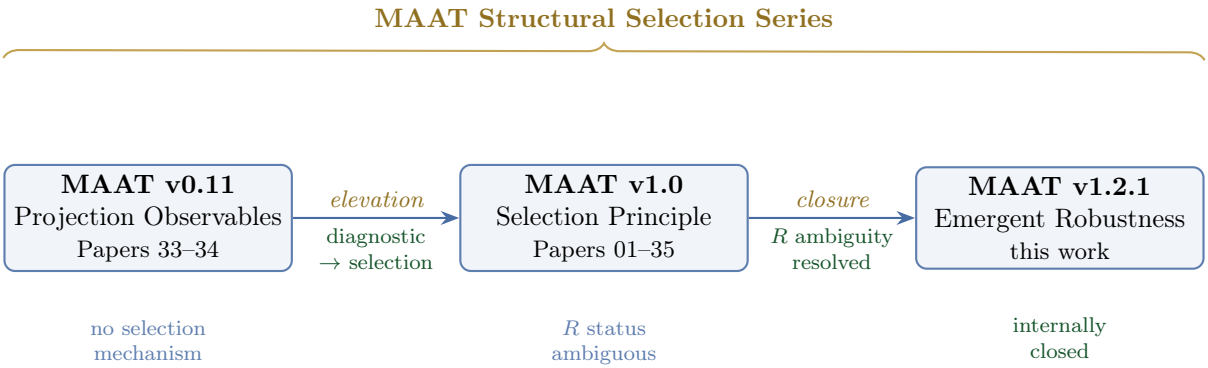
1. **v0.11** — Projection observables: a new layer of cosmological information is identified and formalised, but the underlying selection mechanism is not yet defined.

2. **v1.0** — Structural selection principle: the structural functional $\mathcal{F}_{\text{MAAT}}$ is elevated from a diagnostic to a selection law, with five explicit sectors (H, B, S, V, R) (v1.0 synthesis based on Papers 01–35).

3. **v1.2.1** — Emergent robustness closure: the ambiguous status of R as an independent sector is resolved by reconstructing it as an emergent property of the other four fields.

The conclusion is that MAAT v1.2.1 is strictly more closed than its predecessors: fewer free parameters, clearer physical interpretation, and the same baseline observable pipeline.

2 Evolution Diagram



3 Stage I: Projection Observables (v0.11)

3.1 The projection layer

MAAT v0.11 introduced a projection layer on top of standard cosmological observables [23, 24]. The central quantity is the projection CCI:

$$C_{\mathrm{proj}}(z) = \frac{S_{\mathrm{proj}}(z)}{1 + \gamma\, D_{\mathrm{acc}}(z)}, \tag{1}$$

where S_{proj} encodes expansion-driven breadth and D_{acc} encodes accessible structural depth. The transition marker

$$z_{\mathrm{tr}} = \arg\max_{z} \left| \frac{d^2 R_{\mathrm{proj}}}{dz^2} \right| \approx 1.044 \quad \text{(v0.11 residual)} \tag{2}$$

identifies the redshift at which projection stress is maximal.

3.2 What v0.11 established

v0.11 result. $C_{\mathrm{proj}}(z)$ does not reduce to standard growth observables such as $f\sigma_8$ under one-parameter rescaling ($\chi^2/\mathrm{dof} \approx 42.8$ vs 1.05 for ΛCDM). This confirms that the projection observable encodes a genuinely distinct layer of cosmological information.

3.3 Limitation of v0.11

The projection layer is diagnostic. It identifies and quantifies a structural observable, but it does not define a fundamental selection mechanism. The question *why* certain configurations are selected over others remains unanswered at this stage.

4 Stage II: Structural Selection Principle (v1.0)

4.1 The selection functional

The diagnostic gap of v0.11 is closed by elevating the structural functional to a selection law [18, 19]:

$$\boxed{\mathcal{F}_{\mathrm{MAAT}}[X] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[X]), \qquad P[X] \propto \exp\!\big[-\mathcal{F}_{\mathrm{MAAT}}[X]\big].} \tag{3}$$

This step transforms MAAT from a descriptive framework into a candidate effective theory. Equation (3) is derived, not postulated: it arises as the least-biased probability measure over admissible configurations subject to fixed structural constraints (maximum relative entropy principle [1, 3], formalised in Paper 24 via large-deviation theory [4, 5]).

4.2 The five sectors

v1.0 introduces five structural sectors with explicit physical definitions:

Sector	Symbol	Physical meaning
Harmony	H	Equation-of-motion consistency
Balance	B	Energy-momentum conservation
Creative Structure	S	Controlled dynamical activity
Connectedness	V	Coupling and coherence
Respect / robustness input	R	v1.0 robustness and constraint satisfaction

The stability formula is:

$$\text{Stability} = \min\Big(R,\ (HBSV)^{1/4}\Big)\,. \tag{4}$$

4.3 Cosmological pipeline (v1.0)

The v1.0 selection functional connects to observables through the pipeline established in Paper 31 [22]:

$$d_a \ \to\ C_{ab} \ \to\ \lambda_a(t) \ \to\ \lambda_a(x,t) \ \to\ T^{\text{MAAT}}_{\mu\nu} \ \to\ \text{FLRW} \ \to\ H(z),\ \Omega(z),\ f\sigma_8(z). \tag{5}$$

Key v1.0 cosmological results (all MAAT parameters subdominant relative to ΛCDM values from Planck [8]):

$$\Omega_{\text{MAAT},0} = 0.00331, \qquad w_{\text{MAAT},0} = -0.801, \qquad \max|\Delta H/H| = 3.99\%. \tag{6}$$

The equation-of-state parameterisation follows the CPL convention [12]. Growth-rate proxies use the standard $f \approx \Omega_m^{\gamma}$ approximation with $\gamma = 0.55$ [10].

4.4 Limitation of v1.0: ambiguity of R

Conceptual tension in v1.0. While H, B, S, V have clear physical definitions as defect residuals, the status of the v1.0 input R is ambiguous: Is R a fundamental independent field? A fitted parameter? Or does it emerge from the other sectors? The calibration of Paper 27 [20] finds $R \approx 39.17\%$ of total weight — making it the dominant sector — but does not resolve whether this dominance is fundamental or derived.

5 Stage III: Emergent Robustness Closure (v1.2.1)

5.1 Resolution of the R ambiguity

MAAT v1.2.1 resolves the ambiguity by reconstructing R from the other sectors (companion stability analysis, [26]). The response-based closure of the weights λ_a follows the Onsager–Kubo response-theory tradition [6, 7], applied to structural defect covariance. The closure acts on *normalised support scores*, not on raw defect densities or unnormalised physical residuals. If $d_a[\Phi] \geq 0$ denotes the primitive defect in sector a, the corresponding support is

$$\Gamma_a[\Phi] = \frac{1}{1 + d_a[\Phi]} \in (0, 1]. \tag{7}$$

For readability, the symbols H, B, S, V in the following equations denote the support scores $\Gamma_H, \Gamma_B, \Gamma_S, \Gamma_V$. Raw defects must first be mapped to supports before entering the geometric means.

The emergent respect measure is:

$$R_{\text{resp}} = (H \cdot B \cdot V)^{1/3}. \tag{8}$$

Emergent robustness is then defined as:

$$\boxed{R_{\text{rob}} = \min\Big[(H \cdot B \cdot V)^{1/3},\ (H \cdot B \cdot S \cdot V)^{1/4}\Big]\,.} \tag{9}$$

The v1.0 stability formula (4) is recovered with $R = R_{\text{resp}}$. The selection functional and observable pipeline remain unchanged; the closure of the robustness sector is modified. Thus, stability is no longer defined using R as an independent input, but is identified directly with the emergent robustness:

$$\text{Stability} = R_{\text{rob}}. \tag{10}$$

5.2 Physical meaning of the closure

Equation (9) encodes two conditions:

1. **Implicit respect**: $(HBV)^{1/3}$ — when harmony, balance, and connectedness are jointly high, structural respect emerges automatically. No separate input is required.
2. **Full geometric mean bound**: $(HBSV)^{1/4}$ — the robustness is additionally bounded by creative structure S. A system that is coherent but inactive is not robustly selected.

The two-sided minimum ensures that robustness is neither too optimistic (constrained by S) nor too conservative (lifted by V).

5.3 Comparison v1.0 vs v1.2.1

Quantity	v1.0	v1.2.1
Primary input fields	H, B, S, V, R (5)	H, B, S, V (4)
Respect	independent input	$R_{\text{resp}} = (HBV)^{1/3}$
Robustness	$\min(R, (HBSV)^{1/4})$	$R_{\text{rob}} = \min(R_{\text{resp}}, (HBSV)^{1/4})$
Observable pipeline	—	unchanged
Numerical predictions	—	may shift on recalibration
CCI diagnostic	original R input	emergent R closure
Free parameters	5 sector weights	4 primary + closure
Causal reading	R as hard constraint	R as emergent coherence

5.4 The MAAT v1.2.1 Master Equation

The full master equation of MAAT v1.2.1 combines the selection functional with the emergent robustness closure:

$$\boxed{P[\Phi] \propto \exp(-\mathcal{F}_{\text{MAAT}}[\Phi])\,, \quad \mathcal{F}_{\text{MAAT}}[\Phi] = -\sum_{a\in\{H,B,S,V\}} \lambda_a \log(\epsilon + \Gamma_a[\Phi]) - \lambda_{\text{cl}} \log(\epsilon + R_{\text{rob}}[\Phi]),} \tag{11}$$

where R_{rob} is *derived*, not a primary field. The four primary fields (H, B, S, V) carry independent weights λ_a; R_{rob} enters through the closure coefficient λ_{cl}. In the canonical v1.2.1 formulation, λ_{cl} is not an independent primary weight but is derived from the covariance/response

structure of the four primary sectors. Paper 30 [21] provides the first effective derivation of sector weights as response pressures:

$$\lambda = \left(C + \eta \, \frac{\mathrm{tr}(C)}{A} \, I \right)^{-1} \left(\langle d \rangle_{\mu_0} - d^* \right), \tag{12}$$

where $C = \mathrm{Cov}_{\mu_0}(d_a, d_b)$ is the defect covariance matrix, d^* is the target defect vector, and η is a ridge regularisation parameter. The closure coefficient λ_{cl} follows the same logic applied to the derived sector R_{rob}. It may be calibrated empirically, but it is conceptually a derived effective parameter, not a fifth free sector weight.

The structural sectors and their meanings are:

Symbol	Formula	Physical meaning
d_a	primitive defect	raw structural violation in sector a
Γ_a	$1/(1+d_a)$	normalised support score $\in (0,1]$; high = low defect
λ_a	response-derived	selection pressure conjugate to defect d_a
λ_{cl}	response-derived	closure weight for the R_{rob} term; not primary
ϵ	small const.	numerical regulariser preventing log(0)
Γ_H	$1/(1+d_H)$	equation-of-motion consistency (high = near solution)
Γ_B	$1/(1+d_B)$	energy-momentum conservation (high = conserved)
Γ_S	$1/(1+d_S)$	controlled activity / creative structure (high = active)
Γ_V	$1/(1+d_V)$	connectedness / coherence (high = well-coupled)
R_{resp}	$(\Gamma_H \Gamma_B \Gamma_V)^{1/3}$	implicit respect: constraint adherence
R_{rob}	$\min(R_{\mathrm{resp}}, (\Gamma_H \Gamma_B \Gamma_S \Gamma_V)^{1/4})$	emergent robustness: stability under perturbation
Stability	$= R_{\mathrm{rob}}$	selection stability diagnostic

In equations:

$$\Gamma_a = \frac{1}{1 + d_a[\Phi]} \in (0,1], \quad a \in \{H, B, S, V\}, \tag{13}$$

$$R_{\mathrm{resp}} = (\Gamma_H \cdot \Gamma_B \cdot \Gamma_V)^{1/3}, \tag{14}$$

$$R_{\mathrm{rob}} = \min\Big[(\Gamma_H \cdot \Gamma_B \cdot \Gamma_V)^{1/3}, \; (\Gamma_H \cdot \Gamma_B \cdot \Gamma_S \cdot \Gamma_V)^{1/4} \Big], \tag{15}$$

$$\mathrm{Stability} = R_{\mathrm{rob}}. \tag{16}$$

The cosmological projection adds the observable layer:

$$C_{\mathrm{proj}}(z) = \frac{S_{\mathrm{proj}}(z)}{1 + \gamma \, D_{\mathrm{acc}}(z)}, \qquad z_{\mathrm{tr}} = \arg\max_z \left| \frac{d^2 R_{\mathrm{proj}}}{dz^2} \right| \approx 1.044. \tag{17}$$

Compared to v1.0, the master equation (11) is **identical in form**. What changes is the closure: in v1.0, R was an independent input with its own λ_R; in v1.2.1, R_{rob} is derived from

$(\Gamma_H, \Gamma_B, \Gamma_S, \Gamma_V)$ with no additional primary support field. In the minimal convention, the closure coefficient is response-derived rather than freely introduced.

v1.2.1 pipeline summary.

$$d_a \;\to\; \Gamma_a \;\to\; R_{\text{resp}},\, R_{\text{rob}} \;\to\; \mathcal{F}_{\text{MAAT}} \;\to\; P[\Phi] \;\to\; T^{\text{MAAT}}_{\mu\nu} \;\to\; H(z),\, C_{\text{proj}}(z),\, z_{\text{tr}}.$$

Four primary defects in, one structural probability out.

5.5 Continuity with the Paper 27 calibration

In Paper 27, R carried approximately 39% of total weight — the dominant sector. The v1.2.1 transition must explain this dominance without introducing R as an independent input.

The explanation follows directly from the Paper 27 runtime model:

$$\log_{10}(\text{runtime}) \approx \beta_0 + \beta_\alpha\, \alpha + \beta_V V + \epsilon(H, B, S, R), \tag{18}$$

where V is the primary driver and R appears only in the correction ϵ. Since $R_{\text{resp}} = (HBV)^{1/3}$ is primarily driven by V when $H, B \approx 1$, the emergent formulation naturally recovers V-dominance. The $R \approx 39\%$ finding is not contradicted. This suggests that the earlier R-dominance can be reinterpreted as boundary/connectedness-driven robustness, rather than as evidence of an independent fifth field. It is a structural reinterpretation, not a proof of the emergent closure.

5.6 Emergent vs effective R

The v1.2.1 closure introduces a useful duality:

Level	R status	Application
Theoretical / structural	$R_{\text{resp}} = (HBV)^{1/3}$ emergent	all theory work
Empirical	$R_{\text{eff}} = R_{\text{resp}} + \delta R_{\text{sys}}$	data regression

A hybrid interpolation with parameter $\alpha \in [0, 1]$ is available:

$$R_{\text{hybrid}}(\alpha) = (1 - \alpha)\, R_{\text{resp}} + \alpha\, R_{\text{explicit}}. \tag{19}$$

For theoretical and cosmological proxy work, $\alpha = 0$ is the preferred minimal default. A non-zero α is warranted only for empirical regression tasks where the primary fields leave significant unexplained variance — for example, in systems with topological protection or discrete-symmetry robustness.

5.7 Relation to k-essence and dark energy

The MAAT kinetic scalar field used in the numerical pipeline shares formal similarities with k-essence [11] and dark-energy scalar fields [9]. The key difference is interpretive: in MAAT, the scalar field encodes structural selection pressure rather than a fundamental energy component. The framework is closer in spirit to the effective-field-theory approach to cosmological perturbations [13] than to a completion of gravity.

5.8 Cosmological consistency

The parameter scan of the v1.2.1 stability analysis [26] over $(\eta, \gamma) \in [0, 0.16] \times [0.05, 1.20]$ (1089 points) confirms:

- All 1089 points satisfy internal stability criteria.

- Best model: $\max|\Delta H/H| = 1.70\%$, $\max|\Delta f\sigma_8/f\sigma_8| = 1.83\%$ (proxy approximation; the full growth embedding of Paper 35 [25] gives 0.059% and 0.457% respectively with a true ΛCDM reference).
- Mean emergent robustness: $\langle R_{\rm rob}\rangle = 0.805$.
- Within the scanned two-parameter proxy space, the baseline is not an isolated fine-tuned point.

6 The Unified Progression

The three stages form a closed logical sequence:

$$\underbrace{\text{diagnostics}}_{\text{v0.11}} \longrightarrow \underbrace{\text{selection}}_{\text{v1.0}} \longrightarrow \underbrace{\text{closure}}_{\text{v1.2.1}}. \tag{20}$$

Each stage resolves a specific open problem:

Stage	Open problem	Resolution	Papers
v0.11	No selection mechanism	—	33–34
v1.0	Selects but R is ambiguous	MaxEnt + 5 sectors	01–35
v1.2.1	R ambiguity	R reconstructed	this work

Key insight. The progression from v0.11 to v1.2.1 does not add complexity. It removes it. Each step eliminates an ambiguity and closes an interpretive gap. The framework becomes progressively more constrained and physically interpretable — without modifying the selection functional or observable pipeline.

7 Physical Proxy Validation

The v1.2.1 sector definitions can be tested against a simple controlled system: an electromagnetic field in vacuum.

For a plane wave $F^{\mu\nu}$ satisfying Maxwell's equations:

$$d_H = \frac{\|\partial_\mu F^{\mu\nu}\|^2}{\Lambda_H^2 + \|\partial_\mu F^{\mu\nu}\|^2} = 0 \;\Rightarrow\; H = 1, \tag{21}$$

$$d_B = \frac{\|\partial_\mu T^{\mu\nu}_{\rm EM}\|^2}{\Lambda_B^2 + \|\partial_\mu T^{\mu\nu}_{\rm EM}\|^2} = 0 \;\Rightarrow\; B = 1. \tag{22}$$

$H = 1$ and $B = 1$ are not assumed: they follow from $\partial_\mu F^{\mu\nu} = 0$ and $\partial_\mu T^{\mu\nu}_{\rm EM} = 0$ in vacuum.

With first-order coherence $V = |\gamma^{(1)}|$ and normalised field activity $S = \langle E^2+\mathcal{B}^2\rangle/(\Lambda_S^2+\langle E^2+\mathcal{B}^2\rangle)$, the emergent robustness for coherent laser light ($S = 0.5$ as an illustrative normalisation choice, $V = 0.99$) is:

$$R_{\rm rob} = \min\Big[(1\cdot 1\cdot 0.99)^{1/3},\ (1\cdot 1\cdot 0.50\cdot 0.99)^{1/4}\Big] = \min(0.997,\ 0.839) = 0.839. \tag{23}$$

For ideal coherent light ($S = 1$, $V = 1$): $R_{\rm rob} = 1$. For incoherent or weak light: $R_{\rm rob} \ll 1$. The sector definitions correctly recover physically expected behaviour.

8 What Has Changed and What Has Not

Element	Status	Comment
$\mathcal{F}_{\mathrm{MAAT}}[X]$ (Eq. 3)	**unchanged**	selection functional
Observable pipeline	**unchanged**	$H(z)$, $w(z)$, $f\sigma_8(z)$
Numerical predictions	*may shift*	under recalibration of λ_a
Pipeline $d_a \to \cdots \to$observables	**unchanged**	Paper 31
Stability formula	*reinterpreted*	v1.0: $R = R_{\mathrm{resp}}$; v1.2.1: R_{rob}
R as input	*reinterpreted*	$\to R_{\mathrm{resp}} = (HBV)^{1/3}$
CCI with R	*reinterpreted*	now R_{rob} in denominator
Number of free sectors	reduced	$5 \to 4$

9 Respect vs Robustness: A Conceptual Clarification

The terms *respect* and *robustness* appear closely related in the MAAT framework but refer to distinct structural levels. This section makes the distinction explicit.

9.1 Respect (R_{resp}): structural constraint adherence

Respect is a **qualitative structural property**: the degree to which a configuration adheres to the constraints defined by the other sectors. It is not a force, not an energy, and not a conserved quantity. It is a measure of *structural integrity under constraint*.

In v1.2.1, respect is defined on normalised supports:

$$R_{\mathrm{resp}} = (\Gamma_H \cdot \Gamma_B \cdot \Gamma_V)^{1/3}. \tag{24}$$

In prose we often write this as $(HBV)^{1/3}$, with the convention that H, B, V denote support scores rather than raw defects. Respect is high when H (internal consistency), B (conservation), and V (coupling/coherence) are jointly high. It emerges from the structure — it is not imposed on it.

9.2 Robustness (R_{rob}): emergent stability

Robustness is a **quantitative stability diagnostic**: the degree to which a configuration withstands perturbation without losing its selected status:

$$R_{\mathrm{rob}} = \min\Big[(\Gamma_H \cdot \Gamma_B \cdot \Gamma_V)^{1/3},\ (\Gamma_H \cdot \Gamma_B \cdot \Gamma_S \cdot \Gamma_V)^{1/4}\Big]. \tag{25}$$

Robustness is bounded above by respect and additionally constrained by creative structure S. A configuration can have high respect but low robustness if $S \approx 0$ in a dynamical or generative regime. For strictly static equilibrium sectors, S may be replaced by a domain-specific activity floor or omitted from the stability bound.

9.3 Robustness as a compressed structural invariant

The formula R_{rob} does not introduce new independent components. All aspects commonly associated with robustness are already implicitly encoded in the primary sectors (H, B, S, V):

Robustness aspect	Encoded in
Constraint adherence	$R_{\text{resp}} = (HBV)^{1/3}$
Stability under perturbation	$(HBSV)^{1/4}$
Boundary integrity	H (equation consistency) and B (conservation)
Dynamical resilience	S (controlled activity) and V (connectedness)
Support margin	$\min_a \Gamma_a$ across the active sectors
Perturbation limit	$\min(R_{\text{resp}}, (HBSV)^{1/4})$

The emergent robustness measure is therefore a *compressed structural invariant* rather than a composite of independent sub-variables. No new primary support fields or degrees of freedom are introduced; in the minimal convention the closure coefficient is response-derived.

Central claim. Robustness does not introduce additional independent components. All aspects commonly associated with robustness — stability, resilience, perturbation resistance, and boundary integrity — are implicitly encoded in the structural sectors (H, B, S, V) and their joint geometric constraints.

$$R_{\text{rob}} = \min\left(\underbrace{(HBV)^{1/3}}_{\text{constraint adherence}} \quad , \quad \underbrace{(HBSV)^{1/4}}_{\text{full stability bound}} \right).$$

Robustness is not a new sector — it is the saturation boundary of structural coherence.

9.4 Key distinctions

Property	Respect R_{resp}	Robustness R_{rob}
What it measures	constraint adherence	stability under perturbation
Formula	$(HBV)^{1/3}$	$\min(R_{\text{resp}}, (HBSV)^{1/4})$
Sectors involved	H, B, V	H, B, S, V
Role of S	not included	additional constraint
Relation	—	$R_{\text{rob}} \leq R_{\text{resp}}$ always
Analogy	obeying rules	surviving under pressure

In one sentence. *Respect* asks: does the configuration adhere to its own structural constraints? *Robustness* asks: does it survive perturbation and remain selected?

Robustness is derived from respect, but it is not identical to respect. It is respect constrained by the full four-field stability bound:

$$R_{\text{rob}} = \min(\underbrace{R_{\text{resp}}}_{\text{respect}}, \underbrace{(HBSV)^{1/4}}_{\text{full stability bound}}).$$

A robust configuration must be respectful *and* dynamically active *and* connected. Robustness is the stronger, more demanding condition. Stability = Robustness = min(Respect, full geometric mean).

9.5 Examples

System	R_{resp}	S	R_{rob}	Interpretation
Plane wave (vacuum)	1.00	0.50	0.839	respectful, mod. robust
Ideal laser	1.00	1.00	1.000	fully respectful + robust
Incoherent light	0.70	0.80	0.70	low respect caps robustness
Human (balanced)	0.85	0.75	0.82	respect and activity balanced
Human (conflict)	0.30	0.90	0.30	active but structurally fragile

The human conflict case is instructive: high S (many possible reactions) with low H, B, V gives high activity but low respect — and therefore low robustness. The configuration is energetically active but structurally fragile.

10 Domain-Specific Note on the R_{rob} Closure

The R_{rob} closure is defined at the structural-theory level. Its minimal interpretation is universal:

$$R_{\text{resp}} = (HBV)^{1/3}, \qquad R_{\text{rob}} = \min\Big(R_{\text{resp}}, (HBSV)^{1/4}\Big)\,, \tag{26}$$

where H, B, S, V denote normalised support scores. Domain-specific applications may introduce additional *modulation* or *measurement* layers, but these do not change the ontological status of R_{rob} as a derived closure quantity.

10.1 Compact domain map

Domain	Closure status	Rationale
Physics (fields)	emergent	field equations determine the active supports
Cosmology	emergent	proxy dynamics well-resolved
Human / cognitive	modulated expression	agency and stress may modulate behavioural expression
AI architecture	emergent	consistency, balance, and context-binding define supports
AI safety	hard-gated	explicit constraints can only reduce the effective score
Data regression	effective proxy optional	allowed when primary supports leave residuals

10.2 Modulation and safety gates

For cognitive or behavioural systems, an agency or expression factor $A \in [0, 1]$ may modulate whether structural respect is externally expressed:

$$R_{\text{human}} = R_{\text{resp}} \cdot A, \tag{27}$$

without making respect a new primary field. For AI safety and alignment, hard constraints are better represented as a gate:

$$R_{\text{AI,eff}} = \min(R_{\text{resp}},\ R_{\text{safety}})\,, \tag{28}$$

where $R_{\text{safety}} \in [0, 1]$ represents hard constraints and harm-avoidance rules. This hard-gate form prevents safety from increasing the structural respect score; it can only preserve or reduce the effective score.

Unified interpretation. The R-closure is ontologically emergent in all domains: structural coherence generates respect and robustness as natural consequences. It becomes operationally explicit only when additional measurement, agency, or safety layers modulate how that closure is expressed.

11 Open Questions

The progression from v0.11 to v1.2.1 closes the R-ambiguity. It does not close all open questions:

1. **λ_a physical closure.** The selection weights λ_a are calibrated from defect covariance and target-response geometry. A derivation from microscopic physics (entropy production, RG scale) is not yet established. This is the central open problem of the framework.

2. **CMB/BAO multi-probe fit (next paper).** Paper 32 made first contact with Cosmic Chronometer $H(z)$ data ($\chi^2_{\text{MAAT}} = 15.37$ vs $\chi^2_{\Lambda\text{CDM}} = 14.88$). The next necessary step is a full multi-probe likelihood combining BAO, CMB shift parameters, and supernovae. This constitutes the primary falsifiability test of the framework.

3. **Full perturbation solver.** Current growth calculations use proxy approximations. A Boltzmann-code implementation is required for precision comparison.

4. **Agency extension.** For complex systems with internal degrees of freedom (biological, cognitive), the emergent robustness R_{rob} may require an additional modulation factor $A \in [0, 1]$ representing active structural deviation. This is an open conceptual question.

5. **Independent validation of the emergent R closure.** The relation $R_{\text{resp}} = (HBV)^{1/3}$ should be tested on datasets not used in the boundary-aware calibration of Paper 27. In particular: do the sector correlations from different benchmark classes (optimisation, cosmology, AI alignment) confirm that V-dominance drives the apparent R-dominance consistently?

12 Conclusion

The MAAT framework evolves through three internally consistent stages.

- **v0.11** established that projection observables encode distinct cosmological information not captured by standard growth statistics.
- **v1.0** transformed the diagnostic framework into a structural selection theory, deriving the selection functional from maximum entropy principles.
- **v1.2.1** closed the remaining ambiguity by reconstructing robustness R as an emergent property of coherence, balance, and connectedness.

The result is a more constrained and more interpretable framework with the same baseline observable pipeline. The selection functional and observable pipeline remain unchanged; only the robustness closure changes. The number of independent input sectors decreases from five to four. The former v1.0 R-sector becomes a consequence of structural coherence rather than an independent assumption.

Structural selection is not the addition of a new equation.

It is the recognition that physical realisation is determined by structural coherence — and that robustness is its natural consequence.

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